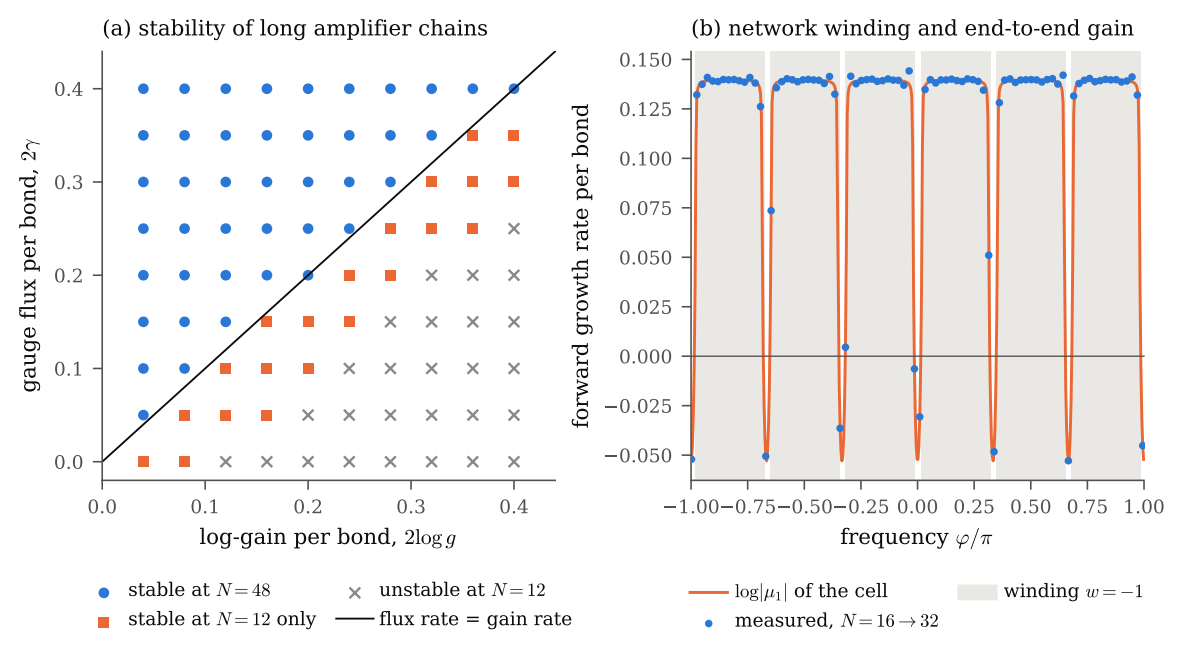
(a) stability of long amplifier chains
gauge flux per bond, 2γ
log-gain per bond, $2\log g$
0.0
0.1
0.2
0.3
0.4
stable at $N=48$
stable at $N=12$ only
unstable at $N=12$
flux rate = gain rate
(b) network winding and end-to-end gain
forward growth rate per bond
frequency φ/π
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
$\log|\mu_1|$ of the cell
measured, $N=16\to32$
winding $w=-1$